

Public Key Crypto notes - 12

By contributors

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1 Elliptic Curve ElGamal Signature Scheme

Let p be a prime, E be an elliptic curve over $\mathbb{Z}_p$, $P \in E$ of prime order l . Also, let $f : E \rightarrow \mathbb{Z}$ (typical choice: $f(x, y) = x$).

- KeyGen:

- $sk = a \in_R \mathbb{Z}_l$, $pk = (p, E, P, l, a \cdot P = Q)$.

- Sign:

- Let $m \in \mathbb{Z}_l$,
 - Choose $r \in_R \mathbb{Z}_l$, $R = rP \in E$,
 - Compute $s = k^{-1}(m - a \cdot f(R))$,
 - Signature: (m, R, s)

- Verify:

- Let $pk = Q$, $sign = (m, R, s)$
 - $v_1 = sR + f(R) \cdot Q$
 - $v_2 = m \cdot P$
 - Verify: $v_1 \stackrel{?}{=} v_2$

- Correctness:

$$v_1 = sR + f(R)Q = k^{-1}(m - af(R))kP + f(R)aP = \quad (1)$$

$$= mP - af(R)P + af(R)P = mP = v_2 \quad (2)$$

2 Elliptic Curve Digital Signature Algorithm (ECDSA)

- Standardized by NIST.

- KeyGen:

- $sk = a \in_R \mathbb{Z}_l$, $pk = Q = aP \in E$

- Sign:

- Let $m \in \mathbb{Z}_l$ be the message.

- Choose $k \in_R \mathbb{Z}_l, R = kP \in E$.
- Compute $s = k^{-1}(m + af(R))$
- Signature: (m, R, s)
- Verify:
 - $u_1 = s^{-1} \cdot m \in \mathbb{Z}_l$
 - $u_2 = s^{-1} \cdot f(R) \in \mathbb{Z}_l$
 - $v = u_1P + u_2Q$
- Correctness:

$$v = u_1P + u_2Q \quad (3)$$

$$= s^{-1} \cdot mP + s^{-1} \cdot f(R)a \cdot P \quad (4)$$

$$= (s^{-1} \cdot m + s^{-1} \cdot f(R)a)P \quad (5)$$

$$= k(m + af(R))^{-1}(m + af(R)) \cdot P \quad (6)$$

$$= kP = R \quad (7)$$

Remark 2.1 (Computational costs of Verify).

EC ElGamal: 3 scalar multiplications, 1 addition.

ECDSA: 2 scalar multiplications, 1 addition on E .

ECDSA is much faster.

Remark 2.2. One can consider elliptic curves over finite field $\mathbb{F}_q$ (q is a prime power) instead of $\mathbb{Z}_p$.

However, if $q = 2^n$ or $q = 3^n$, we need a different curve equation.

3 Pairing-based crypto

Let E be an elliptic curve over $\mathbb{Z}_p$, and for $n \mid |E|$, let $E[n] = \{P \in E, nP = \infty\}$.

If n is large enough, then $E[n]$ can be almost as large as E , e.g. $|E| = l$ is a prime $E[l] = E$.

Definition 3.1 (Pairing). A pairing is a map $e : E[n] \times E[n] \rightarrow \mathbb{Z}_p$ with:

1. e is bilinear. $e(aP, Q) = e(P, aQ) = e(P, Q)^a$
2. e is non-trivial. If $e(P, Q) = 1$ for all Q or $e(Q, P) = 1$ for all $Q \rightarrow P = \infty$

Remark 3.1. For appropriate choice of parameters (e.g. $n \mid p-1$), there are curves E over $\mathbb{Z}_p$ such that a pairing exists.

Having pairing, one can perform cryptographic attacks.

1. If there is a pairing, then the Decisional Diffie-Hellman problem is easy.
 - Given $aP, bP, Q \rightarrow$ decide whether $abP \stackrel{?}{=} Q$
 - Idea: $e(aP, bP) = e(P, abP) \stackrel{?}{=} e(P, Q)$
2. If one can solve DLog in $\mathbb{Z}_p$, then they can do it on E too.

- Given $P, aP = Q \rightarrow a = ?$
- Idea: $e(P, Q) = e(P, aP) = e(P, P)^a$. If $g = e(P, P) \in \mathbb{Z}_p$ given $e(P, Q) = g^a \in \mathbb{Z}_p$.
- So p has to be the same size as in classical (not elliptic curve) crypto, i.e. $p \approx 3000 - 4000$ bits.

3.1 Protocol 1

3-party Diffie-Hellman key exchange protocol.

- Person A : $a \in_R \mathbb{Z}_n$, sends s_AP .
- Person B : $b \in_R \mathbb{Z}_n$, sends s_BP .
- Person C : $c \in_R \mathbb{Z}_n$, sends s_CP .
- Then $e(bP, cP)^a = e(aP, cP)^b, e(aP, bP)^c = e(P, P)^{abc}$.

3.2 Protocol 2 - Identity-based encryption

Goal: Design a PKE such that the public key is an identifier (ID).

Definition 3.2 (Boneh-Franklin IBE).

- A, B users
- TA: Trusted Authority
- Protocols:
 - Setup,
 - Extraction,
 - Encryption,
 - Decryption
- Setup:
 - TA chooses:
 - * Primes p, l , an elliptic curve E over $\mathbb{Z}_p$ such that $\exists e : E[n] \times E[n] \rightarrow \mathbb{Z}_p$ and a point $P \in E[n]$ of order l .
 - * $H : \{0, 1\}^* \rightarrow P \in E[l]$
 - * $s \in_R \mathbb{Z}_l, P_{pub} = sP$, where s is secret, P_{pub} is public.
- Extraction:
 - Let Bob's ID be ID_B .
 - TA computes: $Q_B = H(ID_B), D_B = sQ_B$.
 - Bob's $sk = D_B$
- Encryption:
 - Let ID_B be the ID of Bob.

- A computes: $Q_B = H(ID_B), g_B = e(Q_B, P_{pub})$
- $r \in_R \mathbb{Z}_l$, cipher text: $c = (rP \in E, m \cdot g_B^r \in \mathbb{Z}_l)$ on message m .

• Decryption:

- B is given $c = (U, v)$ having $D_B : h_B = e(D_B, U)$
- $m = \frac{v}{h_B}$

Correctness:

$$h_B = e(D_B, U) = \tag{8}$$

$$= e(sQ_B, rP) = \tag{9}$$

$$= r(Q_B, sP)^r = \tag{10}$$

$$= g_B^r \tag{11}$$

And:

$$\frac{v}{h_B} = \frac{m \cdot g_B^r}{g_B^r} = m$$